Recommended Assessment

Vision

Image Processing

1. In an automated sorting system where the QArm Mini needs to detect and classify objects of different colors under changing lighting conditions, which color space—HSV or RGB—would be more effective?

2. How do the hue, saturation, and value ranges affect object recognition?

3. What is the purpose of applying Gaussian blurring before image transformation? How does changing the kernel size impact detection?

4. How do morphological operations like erosion, dilation, and opening affect image noise and object boundaries? When would you use each?

5. What does the bounding box represent in object detection, and how does changing the connectivity type (4 vs. 8 directions) influence object recognition?

6. What challenges could arise when detecting objects in real-world environments?